S.B. JAIN INSTITUTE OF TECHNOLOGY MANAGEMENT & RESEARCH, NAGPUR

Practical 08

Aim: A system has RAM which can store four pages simultaneously to satisfy page requests. Write a Program to calculate percentage page hit and page fault if the system uses Least recently used page replacement technique for the page references entered by the user i.e. 1, 2, 3, 4, 1, 2, 5, 1, 2, 3, 4, 5 (No of Frames=3).

Name: Shraddha Agalawe

USN: EE24TCD002

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❖ CODE :

```
~
  GNU nano 8.7
#include <stdio.h>

int main() {
    int pages[] = {1,2,3,4,1,2,5,1,2,3,4,5};
    int n = 12;
    int frames[3];
    int time[3];
    int i,j,k,found;
    int hits = 0, faults = 0;
    int counter = 0, pos, min;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {
        found = 0;

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            min = time[0];
            pos = 0;

            for(j=1;j<3;j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            counter++;
            time[pos] = counter;
            faults++;
        }
    }

    printf("Total Page Hits = %d\n", hits);
    printf("Total Page Faults = %d\n", faults);

^G Help          ^O Write Out     ^F Where Is      ^
^X Exit          ^R Read File     ^\ Replace       ^
```

```
~
  GNU nano 8.7
    int time[3];
    int i,j,k,found;
    int hits = 0, faults = 0;
    int counter = 0, pos, min;

    for(i=0;i<3;i++)
        frames[i] = -1;

    for(i=0;i<n;i++) {
        found = 0;

        for(j=0;j<3;j++) {
            if(frames[j] == pages[i]) {
                hits++;
                counter++;
                time[j] = counter;
                found = 1;
                break;
            }
        }

        if(!found) {
            min = time[0];
            pos = 0;

            for(j=1;j<3;j++) {
                if(time[j] < min) {
                    min = time[j];
                    pos = j;
                }
            }

            frames[pos] = pages[i];
            counter++;
            time[pos] = counter;
            faults++;
        }
    }

    printf("Total Page Hits = %d\n", hits);
    printf("Total Page Faults = %d\n", faults);

    printf("Hit Percentage = %.2f%%\n", (hits/(float)n)*100);
    printf("Fault Percentage = %.2f%%\n", (faults/(float)n)*100);

    return 0;
}

^G Help          ^O Write Out     ^F Where Is      ^K Cut
^X Exit          ^R Read File     ^\ Replace       ^U Paste
```

❖ OUTPUT :

```
~

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ nano lru.c

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ gcc lru.c -o lru

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ ./lru
Total Page Hits = 0
Total Page Faults = 12
Hit Percentage = 0.00%
Fault Percentage = 100.00%

Micro_Soft@DESKTOP-QDPORH1 MINGW64 ~
$ |
```